- Allow HTTP access to the proxy from 0.0.0.0/0 or a specific source address.
- Allow HTTP access to all backends only if the traffic source is the proxy.

Figure 5.9 shows that only the proxy is allowed to communicate with the backend over HTTP.

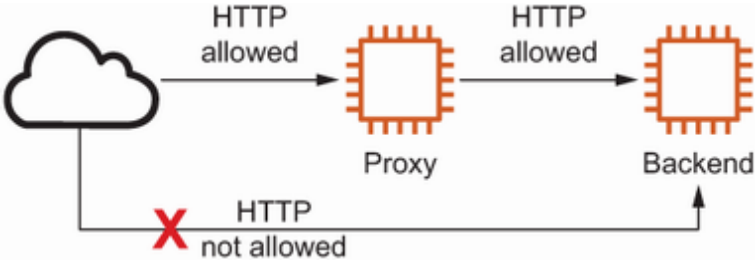


Figure 5.9 The proxy is the only HTTP entry point to the system (realized with security groups).

A security group allowing incoming HTTP traffic from anywhere needs to be attached to the proxy. All backend VMs are attached to a security group allowing HTTP traffic only if the source is the proxy's security group. Listing 5.5 shows the security groups defined in a CloudFormation template.

Listing 5.5 CloudFormation template: HTTP from proxy to backend

```
SecurityGroupProxy:                                          ①
  Type: 'AWS::EC2::SecurityGroup'
  Properties:
    GroupDescription: 'Allowing incoming HTTP and ICPM from anywhere.'
    VpcId: !Ref VPC
    SecurityGroupIngress:
    - Description: 'allowing inbound ICMP traffic'
      IpProtocol: icmp
      FromPort: '-1'
```